

Multilingual LLMs

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

世界需求：大多数人并不以英语为第一语言，MLLM 关系到 fairness、inclusivity 和商业覆盖 → 资源现实：语言资源呈长尾分布，许多语言几乎没有 labelled 或 unlabelled data → 核心张力：共享一个模型能节省成本并产生 positive transfer，但也会带来 negative interference 和 curse of multilinguality → 建模选择：现代模型通常共享 vocabulary、encoder/decoder、attention 等大部分组件，用 prompt 或语言标记 signpost 目标语言 → 数据工程：FineWeb2 代表大规模多语言语料管线，从 language identification 到 tokenisation、deduplication、filtering、rehydration → tokenization 公平：同样一句话在不同语言里可能要花完全不同数量的 tokens，直接影响成本、context window 和模型质量 → 数据混合：natural、uniform/capped、temperature sampling 在高资源语言表现和低资源语言曝光之间做 trade-off → 迁移规律：ATLAS 用 transfer matrix 与 adaptive transfer scaling law 估计哪些语言互相帮助、哪些互相干扰 → 评测责任：不能只报 mean score，也不能迷信机器翻译 benchmark 或 LLM-as-a-judge → Responsible AI：safety alignment 不会自动跨语言迁移，语言社区应参与数据、评测、部署和治理

Promise of MLLMs

- Create AI systems that can operate across languages
 - providing AI for education/healthcare/etc. across the world: **fairness and inclusivity**
 - **commercial potential** - 6.5B people do not speak English
- **More efficient** - training and deploying one model instead of 100s, or even 1000s for cross lingual problems like machine translation
- **Transfer knowledge** between languages – improved performance for low resource languages
- **Deeper understanding** of generative AI models themselves
 - how do they access knowledge and reason across languages?

这节课一句话

Multilingual LLM 不是把英语模型“顺手翻译一下”，而是在数据极度不均衡、tokenizer 不公平、跨语言迁移有利有弊、评测与安全又带文化差异的现实里，设计一个能真正服务多语言社区的模型系统。

1. 为什么 MLLM 是公平问题：资源长尾决定模型命运

Language Resources

Class	5 Example Languages	#Langs	#Speakers	% of Total Langs
0	Dahalo, Warlpiri, Popoloca, Wallisian, Bora	2191	1.2B	88.38%
1	Cherokee, Fijian, Greenlandic, Bhojpuri, Navajo	222	30M	5.49%
2	Zulu, Konkani, Lao, Maltese, Irish	19	5.7M	0.36%
3	Indonesian, Ukrainian, Cebuano, Afrikaans, Hebrew	28	1.8B	4.42%
4	Russian, Hungarian, Vietnamese, Dutch, Korean	18	2.2B	1.07%
5	English, Spanish, German, Japanese, French	7	2.5B	0.28%

88.38% of languages have no labelled or unlabelled data resources!

Joshi et al. (2020) The State and Fate of Linguistic Diversity and Inclusion in the NLP World

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人话直觉

先把这节课想成一个公共服务问题：如果医院问答、教育助手或政府服务 bot 只在英语上靠谱，那它不是“少支持几种语言”，而是在把能力、知识和安全保障集中给已经资源最多的人。更现实的是，英语像高速公路网，很多语言只有碎石小路，甚至地图上都没画出来；模型自然会跑向道路最多的地方。

精确定义 / 机制：Multilingual LLM 的目标是让一个模型在多种语言中理解、生成、推理，并尽可能把高资源语言里学到的知识迁移到低资源语言。共享一个模型比为每种语言单独训练模型更高效，也可能带来 positive transfer。但讲义中最刺眼的背景数字是 88.38% of languages have no labelled or unlabelled data resources。所谓 labelled resources 包括分类、翻译、问答等带标注数据；unlabelled resources 是可用于预训练的大规模文本。资源少并不意味着说话者少或语言价值低，而常常反映数字化、平台权力和研究投入的不平等。数据不均衡会让模型偏向高资源语言，低资源语言更容易出现 wrong language generation、code-switching、事实错误、文化误读和 safety failure。

考试问法：如果考“为什么 multilingual LLM important / challenging”，按 fairness/inclusivity、效率、low-resource transfer、数据长尾四点展开。高分答案会指出：多语言能力不是均匀分布的；low-resource language 的困难来自数据量、数据质量、工具链、benchmark、tokenizer 覆盖和社区参与不足，而不是“语言本身太难”。

2. 建模选择：到底共享什么，怎样告诉模型该说哪种语言

Modelling

- How many components are shared?
 - Many language specific features were proposed: Vocabulary, Encoder, Decoder, Attention Heads
- If shared, need to signpost language:
 - Multilingual machine translation models “En: input sentence” “De: output sentence”
 - “Translate this (Spanish) sentence into English: ”
- Are there multilingual training objectives that help?
- Current models tend to share everything!

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人话直觉

多语言模型像一所国际学校：可以每种语言一个班、一套教材、一批老师；也可以共享老师和教室，但要有清楚的班牌。模型里的“班牌”就是 language tag、prompt instruction 或目标语言说明。

精确定义 / 机制：多语言建模首先问 how many components are shared。早期系统可能为不同语言设计 language-specific vocabulary、encoder、decoder、attention heads 或 embeddings，以减少冲突。现代 LLM 通常倾向于 share everything：同一个 tokenizer、同一套参数、同一组 transformer layers 处理多语言输入。共享带来效率和 transfer，但也需要 signpost language，例如在机器翻译中写“En: input sentence / De: output sentence”，或在 instruction 中明确“Translate this Spanish sentence into English”。XLM/TLM 这类 cross-lingual pretraining 说明，多语言训练目标也能帮助模型形成跨语言表示。

考试问法：如果题目问“what design choices are involved in multilingual modelling”，要答组件共享、语言标记、训练目标、覆盖语言数量。可补一句：current models tend to share everything，但共享不等于无成本，仍会遇到 interference。

3. FineWeb2: 多语言数据集是工程管线，不是网页大杂烩

FineWeb2

Current best practice in building multilingual datasets at scale

- Scale up FineWeb:
 - 100 Common Crawl snapshots spanning the summer of 2013 to April 2024
 - 20 terabyte (5 billion document) dataset
 - covering over 1000 languages
- Quality Filtering:
 - Refined pipeline similar to FineWeb:
Languages Identification -> Tokenisation -> Deduplication -> Filtering -> Rehydration
 - **Rehydration** is new - documents which repeat relatively small number of times eg 10 times are higher quality - single instance often noise, high duplication already overrepresented

Penedo et al (2025) [FineWeb2: One Pipeline to Scale Them All — Adapting Pre-Training Data Processing to Every Language](#)

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人话直觉

训练数据不是“越多越好”的自助餐，而像给模型做食材供应链：先确认食材是什么语言，再切分、去重、过滤，最后判断哪些重复是质量信号，哪些重复是垃圾模板。

精确定义 / 机制： FineWeb2 代表 current best practice in building multilingual datasets at scale：使用 100 个 Common Crawl snapshots，覆盖 2013 年夏天到 2024 年 4 月，形成约 20TB、5B documents、1000+ languages 的大规模语料。它的 pipeline 包括 Language Identification、Tokenisation、Deduplication、Filtering、Rehydration。Rehydration 的直觉很重要：只出现一次的文档可能是噪声；重复太多的文档可能是模板或过度代表；而中等次数重复有时反而说明网页稳定、质量较高。多语言数据尤其需要质量控制，因为 language identification 错误会把样本放进错误语言桶，后续采样和评测都会被污染。

考试问法： 考 FineWeb2 时，按 pipeline 讲最稳：language identification → tokenisation → deduplication → filtering → rehydration。然后解释 rehydration 为什么不是简单去重，而是利用跨 snapshot 的重复模式判断质量。

4. Tokenization 公平：每种语言在模型里付的 token 税不同

Tokenisation – sub-word units

Modelling choice: What is in your vocabulary and how hard is it to represent a text?

Language	ISO	Text	↑ Num Tokens
English	en-US	what will the weather be next week	7
Spanish	es-ES	qué tiempo hará la semana que viene	8
Korean	ko-KR	다음 주 날씨 어때	12
Burmese	my-MM	ဧရာဝတီတစ်ဝတ် ရာသီဥတုဘယ်လိုရှိမလဲ	61
Amharic	am-ET	በሚቀጥለው ሳምንት አየሩ ምን ያህል ነው	69

From [artfish.ai](#)

Tokenizer: cl100k_base tokenizer used in GPT4

Tokenizer choice significantly impacts model performance across languages based on how efficiently it encodes words into tokens

Optimizing for one language often means compromising others.

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人话直觉

同样一句话，有的语言像整齐的硬币，一枚一枚好数；有的语言被 tokenizer 切成一把碎零钱。碎零钱越多，同样意思就越占 context window，也越耗训练和推理预算。

精确定义 / 机制：Subword tokenizer 在字符和词之间折中，但它不是中立工具。Tokenizer choice significantly impacts model performance across languages, 因为它决定每个真实词平均被拆成多少 tokens。FineWeb2 用两个指标看这件事：subword fertility 是每个真实词平均对应几个 subword tokens；proportion of continued words 是有多少真实词被拆成两个或更多 tokens。高 fertility 意味着同样内容更长、context 更快被填满、计算更贵，模型也更难学到稳定词级规律。Gemma tokenizer 的设计包括 SentencePiece/BPE、约 256K 大词表、byte-level encoding、digit splitting 和 preserved whitespace，目标是在多语言覆盖、任意 Unicode 表示和成本之间折中。

subword fertility = $\frac{\text{\#subword tokens}}{\text{\#real words}}$, pcw = $\frac{\text{\#words split into } \geq 2 \text{ tokens}}{\text{\#real words}}$

考试问法：如果考“tokenizer 如何影响 multilingual performance”，要明确写出：它影响 encoding efficiency、context length、training/inference cost 和 representation quality。不要只说 tokenizer 是预处理步骤。

5. Data mix: 采样比例是在英语优势和低资源曝光之间调旋钮

Data Mix

How to balance training data across languages to maximize model performance?

- Trade-off: High-resource languages (abundant data) vs. low-resource languages (limited data)
- Natural distribution heavily favors English (~90% of web data) but underserves other languages

Common Sampling Approaches:

- **Natural/Proportional Sampling** - Use data as available; maintains English dominance
- **Uniform/Capped Sampling** - Cap high-resource languages to prevent over-representation
- **Temperature Sampling** - Up sample low-resource languages by raising proportions to power where:
 p_i is sampling prob for language i , n_i is the natural data proportion, α is the temperature parameters, L is number of languages,

$$p_i = \frac{n_i^\alpha}{\sum_{j=1}^L n_j^\alpha}$$

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人话直觉

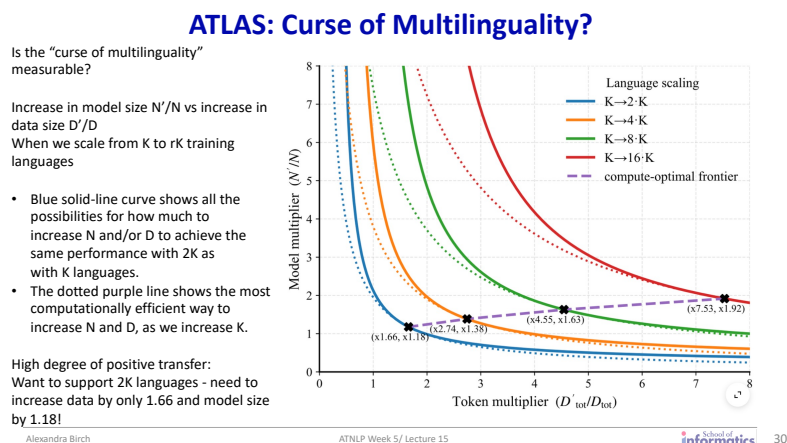
如果完全按网页数量抽样，英语会像自助餐里占满盘子的主菜，低资源语言只能沾一点边；如果每种语言平均抽样，小语种又会被反复吃到过拟合。Temperature sampling 就是在这两个极端之间调火候。

精确定义 / 机制： Data mix 问的是训练时不同语言以什么比例进入模型。Natural/proportional sampling 按自然数据量采样，简单但会维持 English dominance。Uniform/capped sampling 限制高资源语言、提高低资源语言曝光，但可能过度重复小语种。Temperature sampling 用自然比例 n_i 的幂来调节语言 i 的采样概率 p_i ：alpha 越接近 1，越像 proportional；alpha 越小，越接近 uniform。它不是魔法，而是把 high-resource performance 与 low-resource coverage 的 trade-off 显式化。

$$p_i = \frac{n_i^\alpha}{\sum_{j=1}^L n_j^\alpha}$$

考试问法：这是很可能直接考公式的点。答案要先写公式，再解释 alpha 的作用，最后指出风险：低资源语言 upsample 太多会放大噪声、重复少量样本并导致 overfitting。

6. ATLAS 与 curse: 跨语言迁移能被测量, 也要付容量成本



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人话直觉

语言之间像同学互相补课：瑞典语可能帮挪威语，印尼语可能帮马来语；但教室、老师和时间是有限的，课程变多以后，每门课分到的容量也可能变薄。ATLAS 要做的，就是把“谁帮谁、帮多少、什么时候反而拖后腿”测出来。

精确定义 / 机制：传统 English scaling laws 如 Chinchilla 主要关心模型大小、数据量和计算量，并不直接处理 language mix。ATLAS 针对 multilingual setting 拟合 adaptive transfer scaling law：目标语言表现不只取决于本语言数据 D ，而取决于 effective data exposure，即目标语言、相似语言和其他语言分别带来的贡献。Transfer matrix 衡量 language A 的训练数据对 language B 是 positive transfer 还是 negative interference。共享脚本、语言家族或高质量多样数据常带来正迁移；但 curse of multilinguality 也真实存在：语言数量增加后，固定模型容量和训练预算被更多语言共享，单个语言可能表现下降。讲义的重要点是这个 curse 可被 scaling 量化，并会被 positive transfer 缓解；在高正迁移条件下，支持 $2K$ languages 可能只需 data 增加约 1.66 倍、model size 增加约 1.18 倍。训练策略上，低预算时 fine-tune massively multilingual checkpoint 往往更划算；compute 和目标语言数据足够时，from-scratch pretraining 才可能超过 fine-tuning。

$$C \approx 6ND$$

考试问法：考 ATLAS 或 curse 时，不要写成“多语言越多越好/越差”。标准答案是：transfer matrix 量化 synergy/interference；effective data exposure 改写 scaling law；positive transfer 能缓解容量竞争，但策略选择仍取决于 compute、模型大小、目标语言数据和已有 checkpoint。

7. Evaluation: 不要让平均分替低资源语言沉默

Evaluation Best Practice

Advice:

- Choose tasks with appropriate difficulty level. For example, HellaSwag was developed in 2019, when GPT-2 was a SOTA model. And yet, it is still being used to evaluate new models.
- Avoid machine-translated benchmarks, especially without expert verification
- Avoid automatic metrics that correlate poorly with human preferences
 - LLMs fail to detect and penalize errors, such as factual inaccuracies, cultural misrepresentations, and the presence of unwanted language
- Benchmarks should be culturally appropriate and adapted to the target language
 - Eg. the translated version of MMLU in EU21 has not been culturally adapted. Language specific variants of MMLU like CMMLU for Chinese or KMMLU for Korean have been adapted to be culturally relevant.

From <https://huggingface.co/blog/catherinernett/multilingual-best-practices>

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人话直觉

多语言评测最危险的幻觉是：表格最后一列 mean score 很高，于是大家鼓掌。但如果某几个低资源语言分数塌了，平均分就像把警报声调成了背景音乐。

精确定义 / 机制： Multilingual evaluation 不能只把英语 benchmark 机器翻译成别的语言。机器翻译可能不自然、不等价、带英语文化假设，甚至引入错误。Best practice 包括：选择合适难度的任务，避免未经专家验证的 machine-translated benchmarks，谨慎使用与人类偏好相关性差的 automatic metrics，尤其是开放生成任务；benchmark 应 culturally appropriate and adapted to target language。报告结果时不要只报 mean scores，而要按 language、language family、resource level 或 script 分开报告，展示方差和失败案例。

考试问法：考“multilingual evaluation principles”时，用四句答：不要只看平均分；避免未经验证的机器翻译 benchmark；报告 per-language performance；做文化适配和专家验证。可补充 LLM-as-a-judge 在事实性、文化误读和 unwanted language 上不可靠。

8. Responsible AI: 安全、文化和社区不能靠英语迁移自动解决

Centering Linguistic Communities

- Language is not just data—it is cultural knowledge and identity
- Language technology is inherently political and tied to power structures. Blogett et al.
- **Epistemic injustice** is when certain groups' knowledge is excluded or misrepresented.
- Language data often comes from: social media, religious texts, indigenous oral traditions without consent or benefit-sharing.

Multilingual LLMs without communities are **technically multilingual but socially colonial**.
Working with communities shifts AI from **extractive technology** to **participatory knowledge infrastructure**.

Good practice:

Participatory design – Technology should be co-designed with affected users: *defining goals, datasets, evaluation, and deployment*.

Data sovereignty and governance frameworks - Communities control how language data is collected, stored, and used.

Decolonial dataset creation and annotation - Local annotators, paid fairly, including dialects and code-switching

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人话直觉

如果模型的安全护栏主要用英语训练，它就像只认识英文路牌的交警：在英语路口很可靠，到了低资源语言和本地文化语境里，可能看不懂危险信号，也看不懂当地规则。

精确定义 / 机制： Multilingual safety 的核心发现是 safety mechanisms designed for English do not automatically transfer. 非英语，尤其低资源语言中的 malicious prompts 更容易诱发 unsafe responses, 这叫 safety performance gap; 而 general linguistic capability 能迁移，并不代表 safety alignment 也稳健迁移，这叫 fragile safety transfer。讲义还区分 global harms 与 local harms: 前者跨文化普遍有害，后者绑定特定历史、政治、宗教或社区语境。工程上可以用“think in English then respond in original language”暂时降低一些 unsafe responses, 但这仍是英语中心的补丁。更根本的是 centering linguistic communities: participatory design、data sovereignty、decolonial dataset creation、local annotators、dialects 和 code-switching。Masakhane 展示了社区治理、能力建设和 Afro-centric AI design 的替代路径。

考试问法： 如果考 responsible multilingual AI，答案要把 safety 和 community 连起来：安全不能只靠英语对齐迁移；本地 harm 需要本地知识；语言数据涉及身份、权力和治理；好的系统要让受影响社区参与目标、数据、评测和部署。

考试速记版

- MLLM 的价值：fairness/inclusivity、效率、low-resource transfer、理解跨语言 reasoning。
- 88.38% 语言缺少 labelled 或 unlabelled resources，资源长尾是根本背景。
- Positive transfer 是一种语言帮助另一种语言；negative interference 是共享训练造成损害。
- 现代 multilingual LLM 往往 share everything，但需要 language tag 或 prompt signposting。
- FineWeb2 pipeline: language identification → tokenisation → deduplication → filtering → rehydration。
- Tokenizer 不公平会让某些语言 subword fertility 更高，context 和计算成本更贵。
- Temperature sampling 通过 alpha 在 proportional sampling 和 uniform sampling 之间折中。

- Data quality 噪声包括 wrong language、incorrect translation、not language，低资源语言受伤更重。
- ATLAS 用 transfer matrix 衡量语言间 synergy/interference，并改写 multilingual scaling law。
- 不要只报告 mean score；per-language reporting 才能暴露低资源语言失败。
- Machine-translated benchmark 需要专家验证和文化适配，LLM-as-a-judge 也不能盲信。
- Safety alignment 不会自动跨语言迁移；responsible MLLM 必须包含 community participation 和 data sovereignty。

一段稳妥考场回答

Multilingual LLMs are important because they make language technology accessible beyond English, but their central challenge is that languages do not enter training on equal terms. Most languages have little labelled or unlabelled data, so a shared model tends to favour high-resource languages unless data collection, tokenisation and sampling are carefully designed. Key modelling choices include which components to share, how to signpost the target language, and how to exploit positive transfer while avoiding negative interference or the curse of multilinguality. Data pipelines such as FineWeb2 use language identification, tokenisation, deduplication, filtering and rehydration to improve multilingual web data quality. Tokenisation is also a fairness issue: high subword fertility makes some languages more expensive in context and compute. Data mix methods such as temperature sampling balance high-resource performance against low-resource exposure. Evaluation must report per-language results, avoid unverified machine-translated benchmarks, and use culturally adapted tests. Finally, safety and alignment do not automatically transfer from English to other languages, so responsible multilingual AI requires target-language safety work, local harm awareness, participatory design and data sovereignty for language communities.